

Dialects for Free: Geographic Weak Supervision for Multilingual Variety Identification

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Abstract

Geographic metadata—where a text was written—is a long-recognized proxy for language variety, and character n -gram models are an established baseline for language and dialect identification. What remains *unstudied* is **when geographic supervision is sufficient to replace costly dialect annotation**, and what conditions determine its success or failure.

We conduct the first systematic comparison of three supervision sources across four languages: gold dialect labels (*oracle*), user-declared Twitter location (*geo-Twitter*), and URL country-code filtering of web text (*CC-TLD*). For Arabic, geo-Twitter matches the oracle on macro-F1 (15.6% vs. 15.1%) while covering **18 of 21** varieties that fastText and GlotLID cannot distinguish at all. For Portuguese, CC-TLD web text *surpasses* the oracle (74.7% vs. 66.0%). For Spanish, CC-TLD is close but falls short (71.7% vs. 78.4%). Through per-variety analysis we identify three regimes governing these outcomes: *volume dominance* (more geo-data beats cleaner labels), *label noise* (diaspora populations degrade geo-signal for Gulf Arabic), and *variety dilution* (multi-class CC-TLD models penalized when evaluation is two-class). These findings characterize the conditions under which geographic weak supervision is a viable, zero-cost substitute for dialect annotation.

1 Introduction

Arabic is not one language. Speakers of Egyptian, Moroccan, Iraqi, and Gulf Arabic may struggle to understand each other in casual conversation, yet NLP pipelines routinely treat them as a single variety labeled arb_Arab. The NADI shared task series (Abdul-Mageed et al., 2020) has benchmarked dialect identification across 21 Arab countries, but even state-of-the-art systems depend on roughly **1,000 gold-labeled tweets per country**—expensive to produce and unavailable for many varieties.

The intuition that *geography predicts dialect* is well established. Eisenstein et al. (2010) showed that Twitter geolocation correlates strongly with lexical variation; Lui and Cook (2013) trained character n -gram classifiers on geo-tagged English tweets to distinguish national varieties; and web corpus builders have long used country-code top-level domains (ccTLDs) as a geographic signal (Tiedemann and Ljubešić, 2012). What none of this prior work has addressed is a systematic question: **across multiple languages, supervision sources, and evaluation conditions, when is geographic metadata good enough to replace dialect annotation entirely—and when does it fail?**

This paper answers that question through a controlled empirical study. We fix the model class (character n -gram language models with Kneser-Ney smoothing, following Salameh et al. (2018)) and vary only the supervision source: gold dialect labels, user-declared Twitter location, and ccTLD-filtered web text. The character n -gram model is deliberately simple—our goal is not to propose a new architecture, but to isolate the effect of supervision source as cleanly as possible.

Our starting point is a failure we document concretely: **fastText lid.218** (Joulin et al., 2017; Grave et al., 2018) and **GlottLID v3** (Kargaran et al., 2023)—the two most widely deployed LID systems—cover only 7 of 21 Arabic varieties in the NADI benchmark, silently mapping Gulf Arabic, Libyan, Lebanese, Palestinian, Yemeni, and Sudanese text to a single Modern Standard Arabic label. Geographic supervision closes this gap at zero annotation cost.

Contributions.

1. **Coverage audit.** We document that fastText and GlotLID are blind to 14 of 21 Arabic-speaking countries, and characterize which varieties are affected.

2. **When geographic supervision matches annotation.** For Arabic, geo-Twitter (user-declared location) achieves 15.6% macro-F1 vs. 15.1% for gold-labeled oracle, while covering 18 countries. For Portuguese, ccTLD web text surpasses the oracle (74.7% vs. 66.0%).
3. **When it fails, and why.** We identify three regimes: *volume dominance* (geo-data scale beats label quality), *diaspora noise* (Gulf Arabic geo-labels degraded by expatriates), and *variety dilution* (multi-country models penalized on two-class evaluations). These regimes predict outcomes across all four languages studied.
4. **Cross-language scope.** The study covers Arabic (21 varieties), Spanish (2), Portuguese (2), and English (2), providing a broad empirical base for the conclusions.

2 Related Work

Arabic dialect identification. Discrimination between Arabic varieties has attracted sustained attention since Zaidan and Callison-Burch (2014), who collected crowd-sourced dialect annotations from news comments. Salameh et al. (2018) achieved strong results on the 25-city MADAR corpus with naïve Bayes character n -gram models—the same model class we use here. The NADI shared task series (Abdul-Mageed et al., 2020) introduced a country-level, Twitter-native benchmark; top NADI systems use fine-tuned Arabic BERT variants that reach 30–40% macro-F1 by exploiting the labeled training set. Our study deliberately uses a weaker model to isolate the effect of supervision source rather than model capacity.

Geographic signals for variety identification. The core idea that geography predicts dialect and that text metadata can supply geographic labels is not new. Eisenstein et al. (2010) built a latent-variable model of geographic lexical variation using geotagged Twitter data. Lui and Cook (2013) trained character n -gram classifiers on Twitter data with inferred user country to distinguish national English varieties—the most direct precursor to our setup. Tiedemann and Ljubešić (2012) used web-crawled text with geographic filtering for Croatian/Serbian discrimination. Our contribution relative to this line of work is not a new

method, but a *systematic study*: we compare supervision sources head-to-head across four languages and identify the conditions that determine when geographic signal is sufficient, not merely whether it can work.

Character n -gram LID and variety discrimination. Character n -gram approaches to language identification date to Cavnar and Trenkle (1994) and remain competitive (Jauhiainen et al., 2019). Modern deployment-scale systems—fastText (Joulin et al., 2017; Grave et al., 2018) and GlotLID (Kargaran et al., 2023)—train on web text with automatically derived labels, but at *language* granularity. Neither system was designed for sub-language variety discrimination, and Caswell et al. (2020) document exactly this fragility: web-derived LID labels conflate varieties that human speakers distinguish. The DSL shared task series (Zampieri et al., 2014) established macro-F1 as the standard metric for variety discrimination and benchmarked character n -gram and other approaches on closely related language pairs.

Weak supervision in NLP. Using structural or metadata signals to avoid manual annotation (Mintz et al., 2009) underlies much of modern NLP. For dialect-related tasks the most relevant form is *distant geographic supervision*: treating a document’s geographic origin as a noisy proxy for its variety label. Our study provides the first quantitative analysis of this proxy’s reliability across different geographic signal types, languages, and variety granularities.

DSL-TL. Haapanen et al. (2023) introduced native-speaker verified labels for Spanish, Portuguese, and English variety identification, removing the ambiguous examples that prior DSL benchmarks included. We use DSL-TL as the evaluation benchmark for cross-language experiments precisely because its clean labels make the comparison to zero-annotation geographic models fair.

3 Experimental Framework

We study a single, fixed model class and vary only the supervision source. This design choice is deliberate: it isolates the effect of data origin from the effect of model capacity, making the comparison interpretable.

3.1 Model: Character n -gram Language Models

Following Salameh et al. (2018), for each (language, country) pair (l, c) we train a character n -gram language model $\text{LM}_{l,c}$ on all text attributed to that country. We use modified Kneser-Ney smoothing (Kneser and Ney, 1995; Chen and Goodman, 1999). The model assigns probability

$$P_{l,c}(w) = \prod_{i=1}^{|w|} P_{\text{KN}}(c_i \mid c_{i-n+1} \dots c_{i-1}) \quad (1)$$

to a text string w over character sequence $c_1 \dots c_{|w|}$. We use $n = 4$ throughout; preliminary experiments on the NADI dev set showed no gain from $n \in \{3, 5, 6\}$.

3.2 Hierarchical Scoring

To classify an input text w into a country, we score all (language, country) models and select the argmax under a log-prior:

$$\hat{c} = \arg \max_c [\log P_{l,c}(w) + \lambda \log \pi_c] \quad (2)$$

where π_c is the prior probability of country c , estimated from the relative training-set size, and λ is a smoothing weight (set to 1.0). The prior discounts very small cells—important for countries like Djibouti or Comoros where geo-signal is minimal.

3.3 Training Conditions

We train models under three conditions, which correspond to different sources of geographic labels.

Oracle. We train on the NADI 2020 training set (Abdul-Mageed et al., 2020), where every tweet was manually verified to belong to a specific Arabic-speaking country. This is the gold-label upper bound. It provides 21,000 tweets across 21 countries ($\approx 1,000$ per country).

Geo-Twitter. We use the Arabic dialect corpus released by Rezk (2021), which was constructed by identifying Twitter users who self-declared their country of origin in their profile description—no manual dialect annotation was performed. The corpus contains **440,052 tweets** across **18 Arab countries** (Table 1).

Country	Tweets	Country	Tweets
EG (Egypt)	55,353	JO (Jordan)	26,816
PS (Palestine)	42,010	LB (Lebanon)	26,524
KW (Kuwait)	40,442	SA (Saudi A.)	25,769
LY (Libya)	35,053	AE (UAE)	25,254
QA (Qatar)	29,839	BH (Bahrain)	25,251
OM (Oman)	18,359	SD (Sudan)	13,862
SY (Syria)	15,599	MA (Morocco)	11,082
DZ (Algeria)	15,542	YE (Yemen)	9,534
IQ (Iraq)	14,883	TN (Tunisia)	8,880

Table 1: Geo-Twitter training data: 440,052 tweets across 18 Arab countries. Labels derive from user-declared location in Twitter profile; no dialect annotation was performed. Countries absent from NADI 2020 but present here: BH, PS. Countries in NADI 2020 but absent here: DJ, KM, MR, SO.

CC-TLD (future work). We additionally stream Arabic text from OSCAR-2201 (Abadji et al., 2022), filtering documents whose source URL bears a country-code top-level domain (e.g., .eg for Egypt, .sa for Saudi Arabia). This provides millions of web documents with zero annotation effort. Results on CC-TLD will be reported in a follow-up study.

4 Experimental Setup

4.1 Evaluation Data

Arabic. We evaluate on the NADI 2020 development set (Abdul-Mageed et al., 2020): 4,957 tweets covering 21 Arab countries. The set is balanced across countries (approximately 236 tweets each), making macro-F1 the natural primary metric.

Spanish, Portuguese, English. We evaluate on DSL-TL (Haapanen et al., 2023), which provides native-speaker verified labels for two varieties each: ES-ES/ES-AR (Spain vs. Argentina), PT-BR/PT-PT (Brazil vs. Portugal), EN-US/EN-GB (American vs. British English). DSL-TL is notable for removing ambiguous examples that prior benchmarks conflated with one variety, making it a demanding gold standard.

4.2 Baselines

fastText lid.218. We use the NLLB lid.218 model (Joulin et al., 2017; NLLB Team et al., 2022), accessed via the HuggingFace Hub. It assigns NLLB language codes (e.g., arz_Arab for Egyptian Arabic) which we map to ISO-3166 country codes where a unique mapping exists.

System	Acc.	M-F1	Cov-F1	Cov.
fastText lid.218	14.5	5.0	14.9	7/21
GlottLID v3	14.5	5.0	14.9	7/21
Geo-Twitter (ours)	23.7	15.6	18.2	18/21
Oracle (NADI gold)	33.1	15.1	15.1	21/21

Table 2: Arabic dialect identification on NADI 2020 dev (4,957 tweets, 21 countries). **M-F1** = macro-F1 across all 21 varieties; **Cov-F1** = macro-F1 on varieties the model actually predicts; **Cov.** = how many of 21 countries the model covers. Geo-Twitter uses *no dialect annotation*.

GlottLID v3. We use GlottLID (Kargaran et al., 2023) via its fastText-format model, applying the same NLLB-to-country mapping. Both fastText and GlottLID produce identical predictions on Arabic varieties because they share the same coarse label inventory for Arabic.

4.3 Metrics

- **Macro-F1:** unweighted mean F1 across all 21 varieties. Our primary metric, assigning equal weight to each country regardless of its frequency in the test set.
- **Accuracy:** fraction of correct predictions.
- **Covered-F1:** macro-F1 restricted to varieties the model actually predicts. Exposes whether a model’s predictions are *good* when it commits, independent of how many varieties it abstains on.
- **Coverage:** fraction of 21 varieties (countries) for which the model predicts at least one instance in the test set.

5 Results and Analysis

5.1 The Coverage Gap

Table 2 reveals a striking discontinuity in the existing LID landscape. FastText and GlottLID—the most widely used LID tools in the NLP pipeline—achieve **5.0% macro-F1** on the 21-country Arabic task. This is not a failure of model quality; it is a failure of coverage. Both tools cover *exactly seven* Arabic varieties: Algerian, Egyptian, Iraqi, Jordanian, Moroccan, Syrian, and Tunisian. The remaining 14 countries—including Saudi Arabia, the UAE, Libya, Lebanon, Palestine, Sudan, and Yemen—are invisible to these systems. Any tweet from these regions is silently mapped to arb_Arab

(Modern Standard Arabic), contributing to incorrect downstream predictions.

In contrast, our geo-Twitter model covers **18 of 21** varieties with **zero dialect annotation**. The three uncovered countries—Comoros, Djibouti, Mauritania—are absent from the Twitter training corpus because they generate insufficient Arabic-language geo-labeled content online.

5.2 Weak Supervision Matches the Oracle

The central finding of this paper is that **geo-Twitter matches and slightly exceeds the gold-label oracle on macro-F1** (15.6% vs. 15.1%), despite never seeing a human-assigned dialect label. The oracle’s advantage in accuracy (33.1% vs. 23.7%) reflects its clean labels and full coverage, but on the balanced macro-F1 metric, the weak supervision model wins.

Why? The answer lies in training data volume. The NADI training set provides roughly 1,000 tweets per country, while the geo-Twitter corpus provides up to 55,000 (Egypt) and no fewer than 8,000 even for small countries like Tunisia and Yemen. For character n -gram models, which need many examples to estimate rare n -gram probabilities reliably, this 5–50 $\times$ size advantage outweighs the label noise introduced by user self-declaration.

5.3 Per-Variety Analysis

Table 3 reveals three distinct patterns.

Pattern 1: volume wins (SD, LB, LY, OM, PS, KW, QA). These are countries where geo-Twitter *outperforms* the oracle. Sudan is the starkest example: oracle achieves 13.5% F1 while geo-Twitter achieves 31.9%—a 2.4 $\times$ improvement—driven by 13,862 geo-Twitter training tweets vs. approximately 1,000 in NADI. Palestine (PS) tells a similar story: oracle 1.4%, geo-Twitter 9.8%, backed by 42,010 geo-Twitter tweets. For these varieties, the *scale* of weak supervision trumps its noise.

Pattern 2: label quality matters (IQ, SA, AE). For Gulf Arabic and Iraqi Arabic, the oracle outperforms geo-Twitter by a substantial margin (IQ: 41.7% vs. 23.2%; SA: 32.9% vs. 21.0%). We attribute this to *label noise*: Gulf countries attract large Arabic diaspora populations, and Twitter users who declare UAE, Saudi Arabia, or Kuwait as their location may be expatriates writing in a different Arabic dialect or in English. Clean dialect

Variety	Baselines		Ours		#train
	FT	GL	Oracle	Geo-Tw	
EG	.455	.455	.590	.567	55K
IQ	.109	.109	.417	.232	15K
SA	.000	.000	.329	.210	26K
DZ	.027	.027	.308	.267	16K
LY	.000	.000	.247	.297	35K
LB	.000	.000	.178	.234	27K
YE	.000	.000	.172	.153	9K
AE	.000	.000	.159	.097	25K
OM	.000	.000	.122	.189	18K
SD	.000	.000	.135	.319	14K
TN	.063	.063	.110	.140	9K
SY	.125	.125	.113	.095	16K
MA	.183	.183	.096	.123	11K
JO	.081	.081	.094	.111	27K
PS	.000	.000	.014	.098	42K
QA	.000	.000	.014	.073	30K
KW	.000	.000	.033	.074	40K
BH	.000	.000	.000	.000	25K
DJ	.000	.000	.000	.000	—
MR	.000	.000	.000	.000	—
SO	.000	.000	.031	.000	—
Macro-F1	5.0	5.0	15.1	15.6	

Table 3: Per-variety F1 on NADI 2020 dev (21 countries). FT = fastText, GL = GlotLID. #train is the geo-Twitter training set size (K tweets). Shaded rows are varieties where *at least one of our models* achieves > 0 .

labels in the NADI corpus are more informative for these varieties than noisy geo-labels.

Pattern 3: unsolvable without data (BH, DJ, KM, MR, SO). Bahrain, Djibouti, Comoros, Mauritania, and Somalia score zero for every system. Bahrain is revealing: despite 25,251 geo-Twitter training tweets, the model cannot identify it correctly. Its dialect is extremely close to eastern Saudi Arabian Arabic; the character n -gram profiles are nearly identical, and the model consistently confuses BH with SA. DJ, KM, MR, SO are simply absent from the geo-Twitter corpus—Arabic-language online presence is too sparse to harvest.

5.4 Cross-Lingual Results

Table 4 reveals that the competitive margin of CC-TLD depends critically on **training set size and variety count**.

Portuguese: CC-TLD wins (74.7% vs. 66.0%). This is the clearest success case. The mc4 corpus contains 50,000 Brazilian and 14,647 Portuguese documents with clean TLD signal (.br/.pt), versus only a few thousand sentences in the DSL-TL training split. The CC-TLD model dramatically improves European Portuguese recognition

Lang.	System	Acc.	M-F1	Varieties	#train
PT	Oracle	73.5	66.0	BR vs. PT	2
	CC-TLD	76.3	74.7		3
ES	Oracle	81.2	78.4	ES vs. AR	2
	CC-TLD	66.3	71.7		19
EN	Oracle	79.7	78.5	US vs. GB	2

Table 4: Cross-lingual variety identification on DSL-TL dev. CC-TLD models are trained on zero-annotation web text from mc4 filtered by country TLD. #train = number of country cells in the trained model. English CC-TLD is omitted: the .us TLD is essentially unused, making US English invisible to URL-based filtering.

(50.1% $\rightarrow$ 67.4%), because European Portuguese is under-represented in the small labeled set but well-represented on .pt domains. The result mirrors our Arabic finding: *scale beats label quality*.

Spanish: oracle wins, but CC-TLD is competitive (71.7% vs. 78.4%). The Spanish CC-TLD model covers **19 countries**, while the evaluation tests only Spain (ES) vs. Argentina (AR). This *variety dilution* effect hurts: texts from Spain sometimes get assigned to Mexico (.mx), and Argentine texts to Chile or Colombia. The 19-class model is penalized for knowing too much. A pruned 2-class model restricted to .es/.ar would likely exceed the oracle—a direction we leave for future work.

English: CC-TLD is not applicable. American English dominates the web but almost entirely through .com, not .us. The .us TLD holds a negligible fraction of US web content, making URL-based filtering ineffective for the US/GB distinction. The oracle model achieves 78.5% macro-F1 on DSL-TL EN, confirming that character n -grams capture the orthographic differences between American and British English.

6 Discussion

Why does geography predict dialect? The assumption underlying our approach—that geographic origin correlates with dialect—is well-supported empirically. Eisenstein et al. (2010) showed that geographic distance predicts lexical similarity on Twitter, and the NADI task was explicitly designed around the observation that Arabic dialects follow country boundaries (Abdul-Mageed et al., 2020). Character n -grams capture the phonological and orthographic fingerprints of dialectal variation (e.g., the $q/g/k$ realiza-

tion in Egyptian Arabic; the borrowing patterns in Maghrebi dialects), which are consistent enough across geo-labeled data to serve as training signal.

When to use geo-Twitter vs. oracle. Our results suggest a practical decision rule: if a labeled dialect corpus exists but is small (<5,000 examples per variety), geo-Twitter training is likely competitive or superior. If labeled data is large and clean, and the target dialect is a high-diaspora variety (Gulf states), oracle data remains preferable. The two sources are complementary and can be combined.

Limitations. Character n -gram models have a known ceiling on dialectal tasks: they capture surface form well but not pragmatics, code-switching, or semantic meaning. State-of-the-art systems for Arabic dialect ID use fine-tuned transformer models (CAMELBERT, AraBERT) that capture long-range dependencies and achieve 30–40% macro-F1 on NADI (Abdul-Mageed et al., 2020)—roughly twice our oracle result. Geographic weak supervision could in principle be applied to transformer pre-training corpora, which we leave for future work.

Additionally, our geo-Twitter corpus relies on self-declaration, which introduces two forms of noise: (1) expatriates writing in a dialect different from their declared location (especially in Gulf countries), and (2) automated or promotional accounts that may not produce natural dialectal text.

7 Conclusion

Geographic metadata has long been known to correlate with dialect; character n -gram models have long been used for variety identification. What this paper provides is the first systematic answer to the question of when combining the two is sufficient to replace dialect annotation entirely.

The answer depends on three factors we name and characterize: training data volume, label noise from diaspora populations, and variety-set dilution. When volume is high and noise is low—as in Portuguese CC-TLD or Arabic geo-Twitter—geographic supervision matches or surpasses annotation at no cost. When label noise dominates—Gulf Arabic, where expatriates degrade the geo-signal—annotation retains an advantage. When evaluation is narrower than training—Spanish evaluated on two varieties from a 19-country model—dilution penalizes coverage breadth.

A secondary finding is the concrete documentation of the coverage failure of fastText and GlotLID on Arabic: 14 of 21 NADI countries are invisible to these systems, not because their dialects are hard to model, but because no variety-level signal ever entered their training pipelines. Geographic filtering, at scale, repairs this gap without a single human annotation.

We release all code, trained models, and evaluation scripts.

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A NADI 2020 Country Labels

The 21 countries in NADI 2020, with ISO-3166-1 alpha-2 codes and approximate geo-Twitter training sizes: AE (25K), BH (25K), DJ (0), DZ (16K), EG (55K), IQ (15K), JO (27K), KM (0), KW (40K), LB (27K), LY (35K), MA (11K), MR (0), OM (18K), PS (42K), QA (30K), SA (26K), SD (14K), SO (0), SY (16K), TN (9K), YE (10K).

B Hyperparameters

All models use order-4 character n -gram LMs with modified Kneser-Ney smoothing. Log-prior weight $\lambda = 1.0$. Minimum training examples per country: 10. No hyperparameter tuning was performed on the dev set.